

Physics — Transfer & Solution Book

Each page is one question to try on your own. Only open the hints if you are stuck, and only one at a time: the first tells you what to notice, the second tells you which rule fits, the third gives you the first move and stops there. The full worked answer comes after them.

How to start any of these

Equal distance average speed

name the moving body and fill the start and end states before choosing anything

First equation

name the moving body and declare the origin and positive direction

Multiphase kinematics

name the moving body and declare the origin and positive direction

Path vs chord

name the moving body and declare the origin and positive direction

Relative g cancellation

name the moving body and declare the origin and positive direction

Vertical max height

name the moving body and declare the origin and positive direction

Vt signed area

name the moving body and declare the origin and positive direction

Xt slope

name the moving body and declare the origin and positive direction

Transfer question 1 — Average speed equal distance

How much help this one gets: Supported transfer.

Question

A cyclist travels the first half of a straight route at 40 km/h and the second half at 60 km/h. What is the average speed for the whole route?

- (A) 50 km/h
- (B) 48 km/h
- (C) 52 km/h
- (D) 45 km/h

If you get stuck

Hint 1 — What to Notice

Notice which two instants the situation is asking you to compare. Read it from the item before choosing anything.

Hint 2 — Which Rule Fits

Model: average speed. It holds only while the situation states its assumptions.

Hint 3 — Your First Move

First move: name the moving body and fill the start and end states before choosing anything. Stop there and check it before going further.

Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: average speed. Check each assumption against the situation: the situation states its assumptions.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: verify dimensions; verify total distance total time

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Verify dimensions
- Verify total distance total time

The Answer

(B) 48 km/h

One Thing to Check

Before you commit: check the one property this family always has to satisfy — verify dimensions. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: verify dimensions; verify total distance total time. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on average speed equal distance.

Transfer question 2 — Distance displacement

How much help this one gets: Supported transfer.

Question

A runner goes once around a circular track of radius R , returning to the start. What are the distance travelled and the magnitude of the displacement?

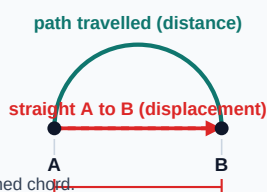
- (A) $2\pi R$ and $2\pi R$
- (B) $2\pi R$ and 0
- (C) 0 and $2\pi R$
- (D) πR and $2R$

If you get stuck

Hint 1 — What to Notice

Notice that a direction has to be declared before any signed quantity means anything. Read it from the item before choosing anything.

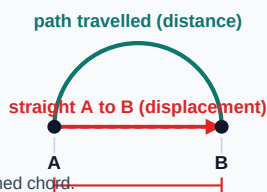
WHAT TO NOTICE FIRST



Hint 2 — Which Rule Fits

Model: particle path geometry. It holds only while the situation states its assumptions.

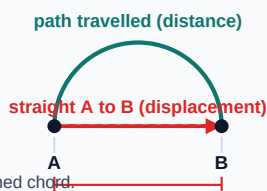
DRAW IT LIKE THIS



Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE



Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: particle path geometry. Check each assumption against the situation: the situation states its assumptions.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: verify dimensions; verify path vs displacement

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Verify dimensions
- Verify path vs displacement

The Answer

(B) $2\pi R$ and 0

One Thing to Check

Before you commit: check the one property this family always has to satisfy — verify dimensions. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: verify dimensions; verify path vs displacement. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on distance displacement.

Transfer question 3 — First equation

How much help this one gets: Independent transfer.

Question

A body starts from rest and accelerates uniformly at 3 m/s^2 . What is its velocity after 6 s ?

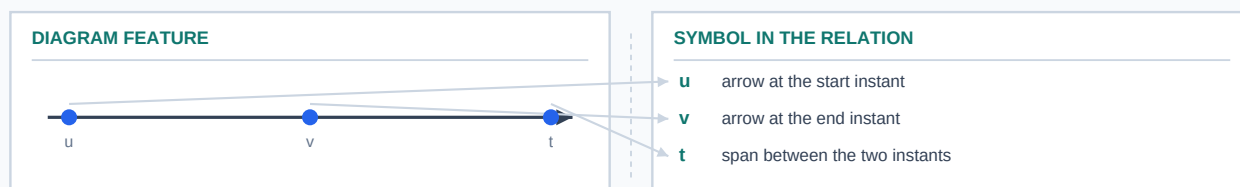
- (A) 9 m/s
- (B) 18 m/s
- (C) 2 m/s
- (D) 36 m/s

If you get stuck

Hint 1 — What to Notice

Notice that a direction has to be declared before any signed quantity means anything. Read it from the item before choosing anything.

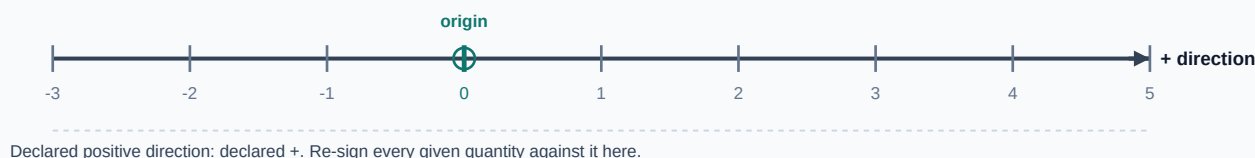
WHAT TO NOTICE FIRST



Hint 2 — Which Rule Fits

Model: uniform acceleration. It holds only while acceleration is constant over the interval.

DRAW IT LIKE THIS



Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE

quantity	start	end
position	?	?
velocity	?	?

Fill both instants first; the empty cell is what the question asks for.

Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: uniform acceleration. Check each assumption against the situation: acceleration is constant over the interval.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: verify dimensions; Check you were allowed to use that rule; verify velocity endpoint

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Verify dimensions
- Check you were allowed to use that rule
- Verify velocity endpoint

The Answer

(B) 18 m/s

One Thing to Check

Before you commit: check the one property this family always has to satisfy — verify dimensions. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: verify dimensions; Check you were allowed to use that rule; verify velocity endpoint. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on first equation.

Transfer question 4 — Vertical projection

How much help this one gets: Independent transfer.

Question

A ball is thrown vertically upward with speed 30 m/s. Taking $g = 10 \text{ m/s}^2$ downward and upward as positive, what is its velocity at the highest point?

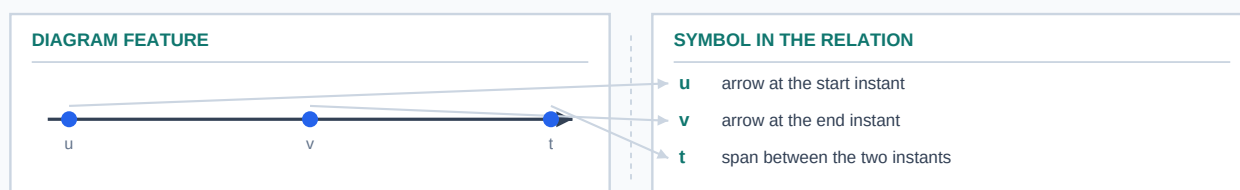
- (A) 30 m/s upward
- (B) 0
- (C) 10 m/s downward
- (D) 30 m/s downward

If you get stuck

Hint 1 — What to Notice

Notice that a direction has to be declared before any signed quantity means anything. Read it from the item before choosing anything.

WHAT TO NOTICE FIRST



Hint 2 — Which Rule Fits

Model: constant g 1d. It holds only while air resistance neglected, g constant.

DRAW IT LIKE THIS



Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE

quantity	start	end
position	?	?
velocity	?	?

Fill both instants first; the empty cell is what the question asks for.

Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: constant g 1d. Check each assumption against the situation: air resistance neglected, g constant.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: verify constant g assumptions; verify dimensions; verify maximum height endpoint

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Verify constant g assumptions
- Verify dimensions
- Verify maximum height endpoint

The Answer

(B) 0

One Thing to Check

Before you commit: check the one property this family always has to satisfy — verify constant g assumptions. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: verify constant g assumptions; verify dimensions; verify maximum height endpoint. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on vertical projection.

Transfer question 5 — Vt signed area

How much help this one gets: Independent transfer.

Question

The velocity-time graph of a body is a straight line from (0 s, 0 m/s) to (4 s, 12 m/s). What is the displacement in the first 4 s?

- (A) 12 m
- (B) 24 m
- (C) 48 m
- (D) 3 m

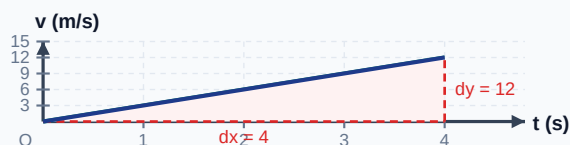
If you get stuck

Hint 1 — What to Notice

Notice what each axis of the given graph actually measures. Read it from the item before choosing anything.

WHAT TO NOTICE FIRST

READ AS SLOPE = a rate



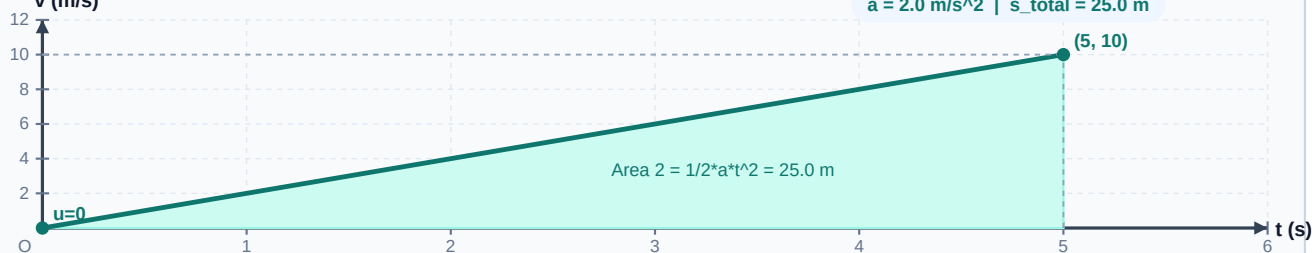
READ AS SIGNED AREA = a change



Hint 2 — Which Rule Fits

Model: graph kinematics. It holds only while the situation states its assumptions.

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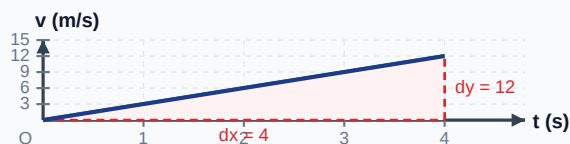


Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE

READ AS SLOPE = a rate



READ AS SIGNED AREA = a change



Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: graph kinematics. Check each assumption against the situation: the situation states its assumptions.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which

variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: verify dimensions; verify signed graph area; verify source representation present

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Verify dimensions
- Verify signed graph area
- Verify source representation present

The Answer

(B) 24 m

One Thing to Check

Before you commit: check the one property this family always has to satisfy — verify dimensions. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: verify dimensions; verify signed graph area; verify source representation present.

A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on vt signed area.

Transfer question 6 — Xt slope velocity

How much help this one gets: Independent transfer.

Question

A position-time graph is a straight line of constant positive slope. What does this tell you about the motion?

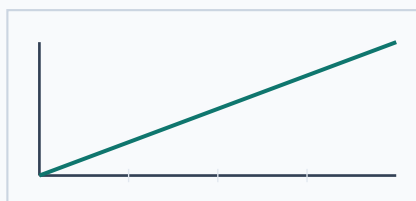
- (A) The body is at rest
- (B) The body moves with constant velocity
- (C) The body accelerates uniformly
- (D) The body reverses direction

If you get stuck

Hint 1 — What to Notice

Notice what each axis of the given graph actually measures. Read it from the item before choosing anything.

WHAT TO NOTICE FIRST

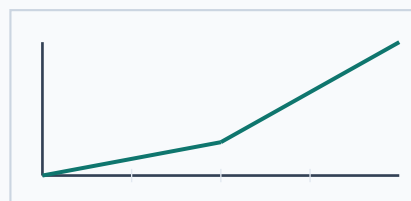


Reject each option by naming the feature it gets wrong.

B



C



Hint 2 — Which Rule Fits

Model: graph kinematics. It holds only while constant positive velocity corresponds to constant positive x-t slope.

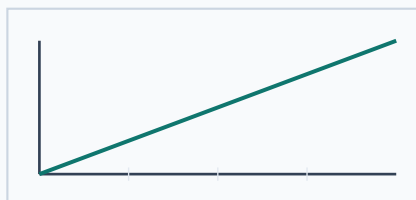
DRAW IT LIKE THIS



Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE

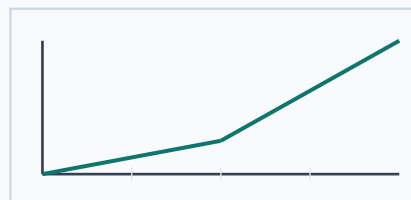


Reject each option by naming the feature it gets wrong.

B



C



Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: graph kinematics. Check each assumption against the situation: constant positive velocity corresponds to constant positive x-t slope.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: verify graph slope sign; verify source representation present

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Verify graph slope sign
- Verify source representation present

The Answer

(B) The body moves with constant velocity

One Thing to Check

Before you commit: check the one property this family always has to satisfy — verify graph slope sign. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: verify graph slope sign; verify source representation present. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on x vs slope velocity.

Transfer question 7 — Multiphase kinematics

How much help this one gets: Independent transfer.

Question

A cart accelerates at 2 m/s^2 for 3 s from rest, then at -1 m/s^2 for a further 2 s . What is its velocity at the end of the 5 s ?

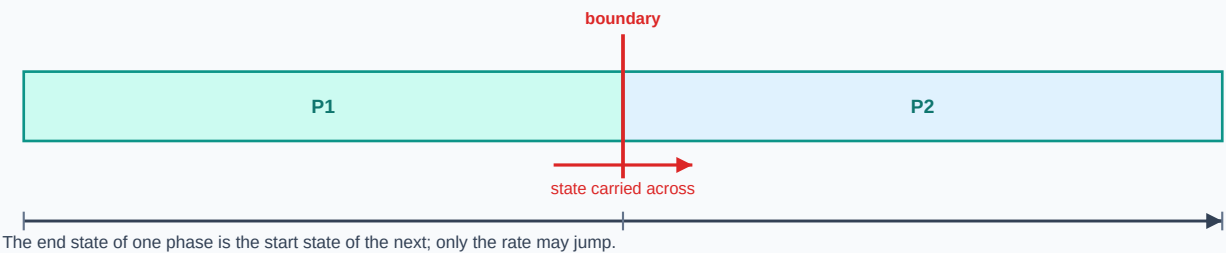
- (A) 6 m/s
- (B) 4 m/s
- (C) 8 m/s
- (D) 2 m/s

If you get stuck

Hint 1 — What to Notice

Notice that the motion does not keep the same character throughout. Read it from the item before choosing anything.

WHAT TO NOTICE FIRST



Hint 2 — Which Rule Fits

Model: piecewise constant acceleration. It holds only while acceleration is constant within each phase, acceleration is constant within each stated phase, phase-boundary state is propagated continuously, velocity and position are continuous at the phase boundary.

DRAW IT LIKE THIS



Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE

quantity	start	end
position	?	?
velocity	?	?

Fill both instants first; the empty cell is what the question asks for.

Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: piecewise constant acceleration. Check each assumption against the situation: acceleration is constant within each phase, acceleration is constant within each stated phase, phase-boundary state is propagated continuously, velocity and position are continuous at the phase boundary.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which

variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: Check you were allowed to use that rule; verify phase state continuity; verify source representation present

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Check you were allowed to use that rule
- Verify phase state continuity
- Verify source representation present

The Answer

(B) 4 m/s

One Thing to Check

Before you commit: check the one property this family always has to satisfy — Check you were allowed to use that rule. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: Check you were allowed to use that rule; verify phase state continuity; verify source representation present. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on multiphase kinematics.

Transfer question 8 — Relative acceleration same g

How much help this one gets: Independent transfer.

Question

Two stones are released from rest at the same instant from different heights. Neglecting air resistance, what is the acceleration of one stone relative to the other?

- (A) g downward
- (B) 0
- (C) $2g$ downward
- (D) g upward

If you get stuck

Hint 1 — What to Notice

Notice that a direction has to be declared before any signed quantity means anything. Read it from the item before choosing anything.

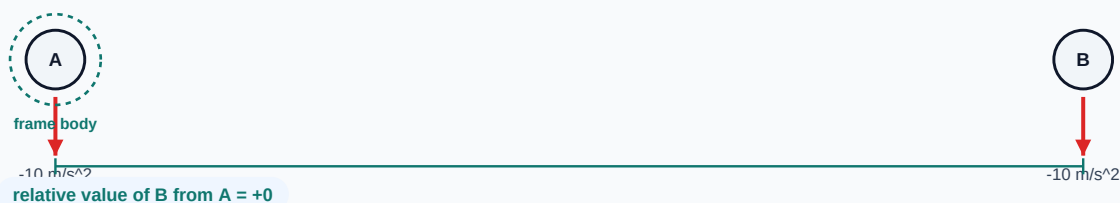
WHAT TO NOTICE FIRST



Hint 2 — Which Rule Fits

Model: relative motion same g . It holds only while same uniform gravitational field.

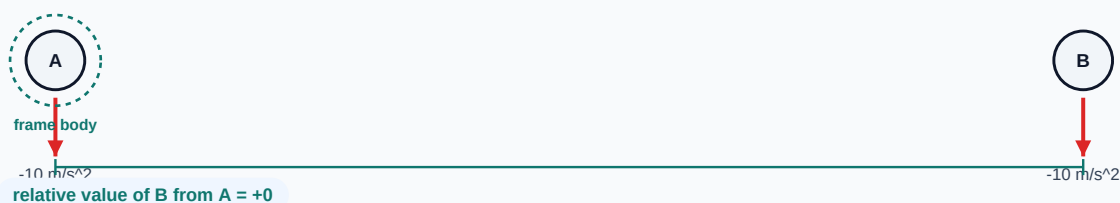
DRAW IT LIKE THIS



Hint 3 — Your First Move

First move: name the moving body and declare the origin and positive direction. Stop there and check it before going further.

YOUR FIRST MOVE — STOP THERE



Full Worked Answer

Set Up Your Directions: Name the body being tracked and declare the origin and positive direction; every signed quantity below uses that one convention.

Write Down What You Know: Enter each stated quantity into the start state or the end state row, with its unit and the sign the declared direction forces, and mark the unknown.

Which Rule Fits Here?: Model: relative motion same g . Check each assumption against the situation: same uniform gravitational field.

Pick the Right Formula: Choose the relation whose variables are exactly the known and wanted state variables, and say which variable it deliberately leaves out.

Do the Working: Substitute the signed values and carry the algebra through without dropping a sign.

Sanity Check: Run an independent check that could reject the answer: Check you were allowed to use that rule; verify relative acceleration cancellation

What the Answer Means: Translate the signed result back into a direction and state what it means physically before selecting an option.

- Check you were allowed to use that rule
- Verify relative acceleration cancellation

The Answer

(B) 0

One Thing to Check

Before you commit: check the one property this family always has to satisfy — Check you were allowed to use that rule. If that fails, the option is wrong whatever the arithmetic said.

Check It a Different Way

Now check it a different way from how you solved it: Check you were allowed to use that rule; verify relative acceleration cancellation. A check that reuses your own working cannot catch your own mistake.

This question practises the same skill taught in the study guide unit on relative acceleration same g .